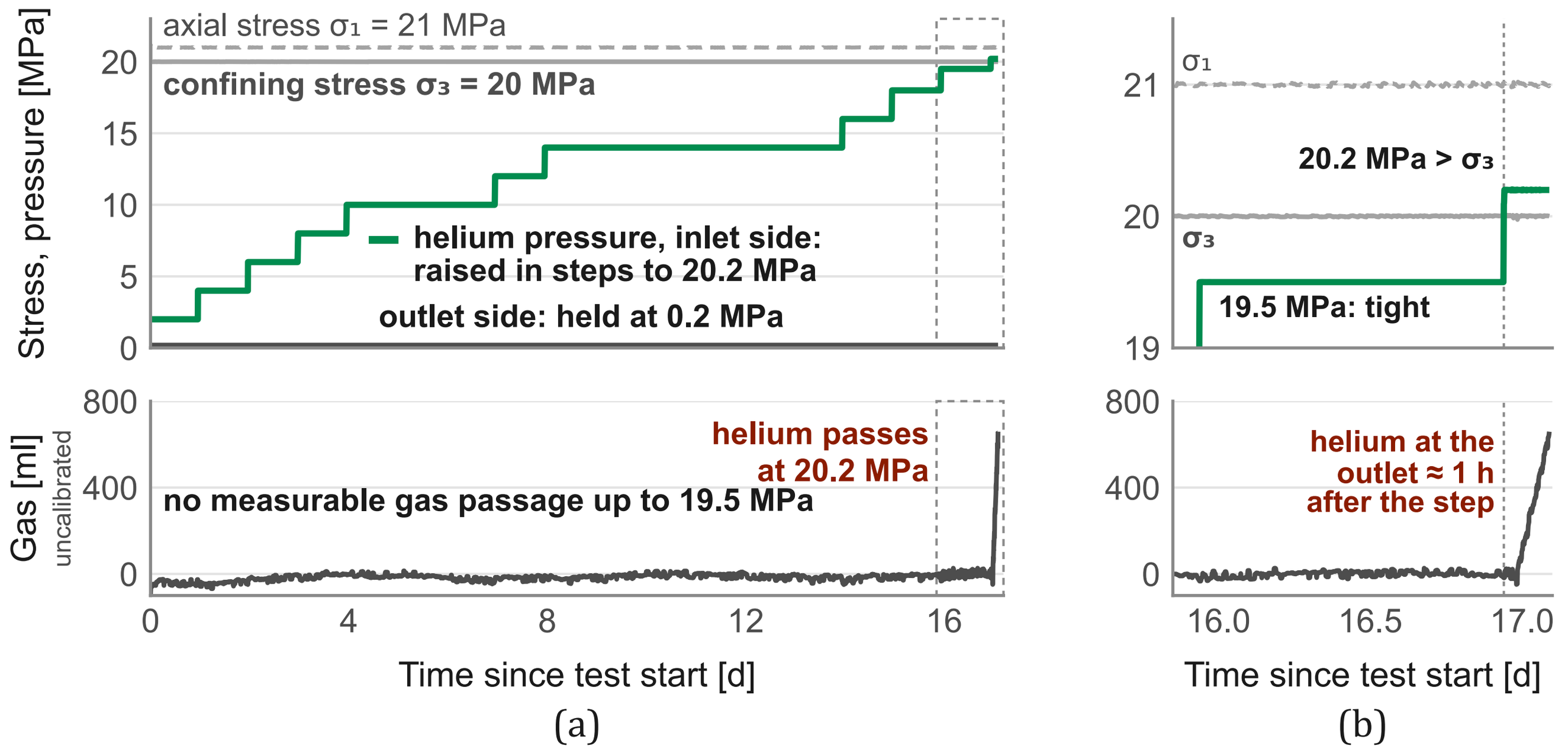

axial stress σ1 = 21 MPa
confining stress σ3 = 20 MPa
helium pressure, inlet side:
raised in steps to 20.2 MPa
outlet side: held at 0.2 MPa
Stress, pressure [MPa]
20
15
10
5
0
helium passes
at 20.2 MPa
no measurable gas passage up to 19.5 MPa
Gas [ml]
uncalibrated
800
400
0
0
4
8
12
16
Time since test start [d]
(a)
σ1
21
20.2 MPa > σ3
20
σ3
19.5 MPa: tight
19
800
400
0
helium at the
outlet ≈ 1 h
after the step
16.0
16.5
17.0
Time since test start [d]
(b)